

Predictions from each neural network architecture for specific urban scene at different input resolutions

Figure 1 shows a specific urban scene with the semantic segmentation ground truth labels.

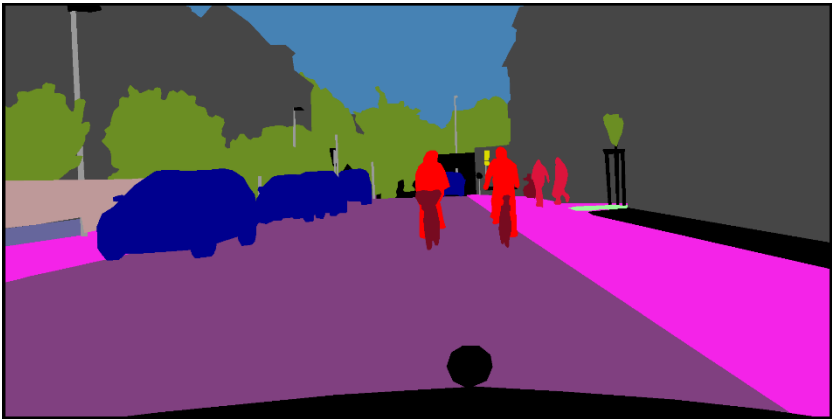
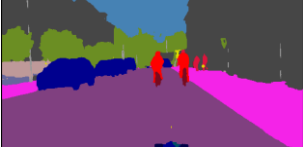
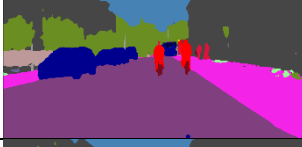
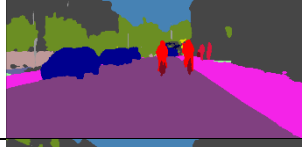
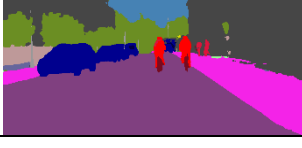


Figure 1. Semantic segmentation ground truth labels for specific urban scene

Table 1 shows the predictions from each neural network architecture for this specific urban scene at different input resolutions.

Table 1. Predictions from each neural network architecture for specific urban scene at different input resolutions

S	U-Net baseline model	Fast-SCNN (CL)	Fast-SCNN (CL + KD)	Fast-SCNN (CL + Dice)
1				
2				
3				
4				
5				

The U-Net model architecture recognizes the bicycles and the leaves of the tree on the right in the first curriculum stage (lowest input resolution).

The Fast-SCNN model architecture does not detect the bicycles and the leaves of the tree on the right in the first curriculum stage (lowest input resolution). These urban elements are recognized in the later curriculum stages (higher input resolutions).

The Fast-SCNN model architecture provides a much better efficiency than the U-Net model architecture at the cost of a reduced but acceptable performance.

With our Fast-SCNN model architecture, we reach the highest efficiency of all TU Eindhoven CodaLab submissions in 2025, and the efficiency is higher than last year’s winning submission. The Fast-SCNN model architecture is superior.

Fast-SCNN balances performance and computational cost, making it particularly suitable for scenarios where both responsiveness and accuracy are critical. With fewer parameters and reduced computation, it is well-suited for mobile and embedded systems.